

(84) Искания канонического
произвольной для группы
независимых переменных

Обобщение произвольной для группы и переменных

$$\begin{cases} y_1 = f_1(x_1, x_2, x_3) \\ y_2 = f_2(x_1, x_2, x_3) \\ y_3 = f_3(x_1, x_2, x_3) \end{cases}$$

$$\mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$\begin{cases} x_1 = \varphi_1(t_1, t_2, t_3) \\ x_2 = \varphi_2(t_1, t_2, t_3) \\ x_3 = \varphi_3(t_1, t_2, t_3) \end{cases}$$

$$\mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$y = f(x(t))$ — сложная группа, композиция

$$\mathbb{R}^3 \xrightarrow{x(t)} \mathbb{R}^3 \xrightarrow{y(x)} \mathbb{R}^3$$

$$\frac{D(y_1, y_2, y_3)}{D(x_1, x_2, x_3)} \cdot \frac{D(x_1, x_2, x_3)}{D(y_1, y_2, y_3)} \stackrel{=}{=} \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix} \cdot \begin{vmatrix} \frac{\partial x_1}{\partial t_1} & \frac{\partial x_1}{\partial t_2} & \frac{\partial x_1}{\partial t_3} \\ \frac{\partial x_2}{\partial t_1} & \frac{\partial x_2}{\partial t_2} & \frac{\partial x_2}{\partial t_3} \\ \frac{\partial x_3}{\partial t_1} & \frac{\partial x_3}{\partial t_2} & \frac{\partial x_3}{\partial t_3} \end{vmatrix} \stackrel{=}{=}$$

$$\stackrel{=}{=} \left| \frac{\partial y_i}{\partial t_k} = \frac{\partial y_i}{\partial x_1} \cdot \frac{\partial x_1}{\partial t_k} + \frac{\partial y_i}{\partial x_2} \cdot \frac{\partial x_2}{\partial t_k} + \frac{\partial y_i}{\partial x_3} \cdot \frac{\partial x_3}{\partial t_k} \right| \stackrel{=}{=} \frac{D(y_1, y_2, y_3)}{D(t_1, t_2, t_3)}$$

$y_i (x_1(t_1, t_2, t_3), x_2(t_1, t_2, t_3), x_3(t_1, t_2, t_3))$

Якобиан - обобщение производной скалярной функции для отображений пространства $\mathbb{R}^n$.

Проверка произвольной обратимой отображением

$$(1); (2.1) \quad \begin{cases} x_1 = \varphi_1(y_1, y_2, y_3) \\ x_2 = \varphi_2(y_1, y_2, y_3) \\ x_3 = \varphi_3(y_1, y_2, y_3) \end{cases}$$

$$\frac{D(y_1, y_2, y_3)}{D(x_1, x_2, x_3)} \cdot \frac{D(x_1, x_2, x_3)}{D(y_1, y_2, y_3)} = \frac{D(y_1, y_2, y_3)}{D(y_1, y_2, y_3)} = \begin{vmatrix} \frac{\partial \varphi_1}{\partial y_1} & \frac{\partial \varphi_1}{\partial y_2} & \frac{\partial \varphi_1}{\partial y_3} \\ \frac{\partial \varphi_2}{\partial y_1} & \frac{\partial \varphi_2}{\partial y_2} & \frac{\partial \varphi_2}{\partial y_3} \\ \frac{\partial \varphi_3}{\partial y_1} & \frac{\partial \varphi_3}{\partial y_2} & \frac{\partial \varphi_3}{\partial y_3} \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$



